Foundation of Special and Inclusive Education

Learning Module No. 02

STUDENT
Name:
Student Number:
Program:
Section:
Home Address:
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LEARNING MODULE INFORMATION	
I. Course Code	
II. Course Title	Foundation of Special and Inclusive Education
III. Module Number	02
IV. Module Title	Children and Youth with Special Education Needs
V. Overview of the Module	This module discusses the children and youth with special education needs. The discussion centers on students with emotional and behavioral disorders, hearing impairment, and visual impairment.
VI. Module Outcomes	At the end of this module, students were able to understand children and youth with special education needs.
VII. General Instructions	You must allot the necessary time to complete the lessons each week. If you choose not to complete the lesson using the schedule provided, you must understand that it is your full responsibility to complete them by the last day of completion. Time is of the essence. The module is designed to assess student understanding of the assigned lessons found within the associated content of the preliminary term, midterm, and final term periods of the course. The assessment part of the module is composed of varied types of questions. You may see true/false, traditional multiple choice, matching, multiple answer, completion, and/or essay. Pay attention to the answer to the assessment questions as you move through each lesson. After each module you will be given a summative test. Your responses to the assessment parts of the module will be checked and recorded. Because the assessment questions are available within the whole completion period and because you can reference the answers to the questions within the content modules, we will not release the answers within modules. However, your professors are happy to discuss the assessments with you during their consultation time, should you have any questions. Good luck. You may not work collaboratively. This is independent work.

Lesson 1: Students with Emotional and Behavioral Disorders

Lesson Objectives:

1. Explain the concepts on personality development, adaptive, and maladaptive behavior;
2. Define the terms emotional and behavioral disorders and delinquency;
3. Enumerate and describe the classification of emotional and behavioral disorders;
4. Enumerate and discuss the characteristics of children with emotional and behavioral disorders;
5. Enumerate and discuss the etiological factors and causes of emotional and behavioral disorders;
6. Describe the assessment tools and procedures in identifying this type of children; and
7. Enumerate and describe the educational approaches for this type of children.

Getting Started (Optional):

"Human behavior flows from three main sources: desire, emotion, and knowledge" - Plato

1) What is your insight or idea with this? Explain.

Discussion:

The variations in human behavior are influenced by the basic determinants of personality development: (1) the person's genetic background or heredity, (2) environmental factors, and (3) the general patterning of development.

Personality development is influenced by a person's socialization experiences. He or she interacts with various modes of behavior patterns that cover a wide range of both positive and negative experiences. His or her behavior patterns reflect much of his or her personality traits which in turn generate either positive or negative responses from the social groups. The social relations vary from acceptance to rejection, inclusion to exclusion and similar reactions.

Adaptive and Maladaptive Behavior

Adaptive behavior refers to a person's behavior patterns that have desirable consequences and foster his or her well-being and ultimately that of the group. The term *well-being* means that the person works towards growth, fulfillment, and actualization of his or her potentials.

Behavior is *maladaptive* if it results to negative and undesirable consequences and interferes with the person's optimal functioning and growth. The use of the term *maladaptive* rather than *abnormal* puts the focus on the behavior rather the person thereby implying the possibility for improvement. Maladaptive behavior includes any behavior that has undesirable consequences for the individual as well as for the group. Behavior problems, emotional and behavioral disorders are manifestations of maladaptive behavior.

The Patterning of Personality Development

1. ***Dependence to self-direction.*** The normal progression is seen in the fetus and newly born infant who are totally dependent on the mother and family members until the toddler begins to explore the environment on his or her own. The child develops into the adolescent and into the independent adult. Bound up with this growth toward independence and self-direction is the development of a clear sense of personality identity and the acquisition of information, competencies, and values.
2. ***Pleasure to reality and self-control.*** The human tendency is to seek pleasure and avoid pain and discomfort. As the child matures, the reality principle takes over and the children realizes that he or she must perceive and face reality. This means distinguishing between fantasy and reality, controlling impulses and desire, delaying immediate gratification in the interest of long-range goals, and learning to cope with the inevitable hurts, disappointments, and frustrations of living.
3. ***Ignorance to knowledge.*** The first two years are characterized by the rapid acquisition of information about themselves and the world. With time, this information is organized into coherent patterns of assumptions concerning reality, possibility, and value that provide a table frame of reference for guiding behavior.
4. ***Incompetence to competence.*** From birth onward to childhood and adolescence, the person masters the intellectual, emotional, social, and other competencies essential for adulthood. The person acquires skills in problem solving and decision-making, learns to control his emotions and to use them for the enrichment of living, and learn to deal with others and to establish satisfying relationships.
5. ***Diffuse to articulated self-identity.*** The core self-identity gradually emerges as the infant differentiates himself or herself from the environment. The sex typing responses of parents and adults produce an awareness of one's self as a boy or a girl. The reactions and feedback from people begin to provide the child with a sense of his or her own characteristics from "good" to "bad".
6. ***Amoral to moral.*** Children learn very early in life that certain forms of behavioral are right, good, correct, while others are bad, wrong, and incorrect to do. As they mature, they learn a pattern of value assumptions that operate as inner guides and controls of behavior which Freud calls the superego or conscience in the psychoanalytic theory of behavior.

Definition of Emotional and Behavioral Disorders

1. *Disordered behavior is social construct. There is no clear agreement as to the criteria and parameters of normal adaptive behavior and good mental health.*
2. *Different theories of emotional disturbance use concepts and terminology that do not present a clear meaning of the condition.*
3. *Measuring and interpreting disorder behavior across time and setting is difficult, exact, and costly endeavor.*
4. *Cultural influence, expectations and norms across ethnic and cultural groups are often quite different.*
5. *Frequency and intensity of disordered behavior is difficult to measure and control since children behave inappropriately at times.*

6. *Disordered behavior occurs in conjunction with other disabilities such as mental retardation and learning disabilities.*

Bower's definition was revised and adopted by the US Department of Education using the term *seriously emotionally disturbed.* Three factors were considered in determining if a child is emotionally disturbed: intensity, pattern, and duration of behavior.

- ***Intensity*** *refers to the severity of the child's problem.*
- ***Pattern*** *means the time when the problem occurs.*
- ***Duration*** *refers to the length of time the child's problem has been present. This implies that continuous observation must be made. Some special educators require three-month duration before they suggest that the child has an emotional or behavioral problem.*

IDEA Definition of Serious Emotional Disturbance (Heward, 2003)

IDEA, the public law on special education in America defines serious emotional disturbance as:

1. A condition exhibiting one or more of the following characteristics over a long period of time (chronicity), and to a marked degree (severity), which adversely affects educational performance (difficulty in school).
   a. An inability to learn which cannot be explained by intellectual, sensory, and health factors;
   b. An inability to build or maintain satisfactory interpersonal relationships with peers and teachers;
   c. Inappropriate types of behavior or feelings under normal circumstances;
   d. A general pervasive mood of unhappiness or depression; or
   e. A tendency to develop physical symptoms or fears associated with personal or school problems.
2. The term includes children who are schizophrenic (autistic). The term does not include children who are socially maladjusted unless it is determined that they are seriously emotionally disturbed.

CCBD Definition of Emotional and Behavioral Disorders (Heward, 2003)

Emotional and behavioral disorders is a disability characterized by:

1. Behavioral or emotional responses in school programs so different from appropriate age, cultural, or ethnic norms that they adversely affect educational performance. Educational performance includes the development and demonstration of academic, social, vocational, and personal skills.
2. Emotional and behavioral disorders can coexist with other disabilities.
3. This category may include children or youth with schizophrenic disorders, affective disorders, anxiety disorders, or other sustained disturbances of conduct or adjustment when they adversely affect educational performance in accordance with section.

Additional factors are considered in assessing a child who is suspected to be emotionally disturbed (Heward, 2003):

- *Rate* refers to the frequency of occurrence of behavior per standard unit of time. Most children cry, get into fights with other children, and sulk from time to time yet they are not emotionally disturbed.
- *Latency* refers to the time that elapses between the opportunity to respond and the beginning of the behavior.

Classification of Emotional and Behavioral Disorders

1. ***The Diagnostic and Statistical Manual of Mental Disorders (DSM – IV)***
   The DSM – IV is an elaborate clinical classification system consisting of 230 separate diagnostic categories or labels to identify the various types of disordered behavior as observed by psychiatrists, psychologists, mental health personnel and other clinicians in their regular practice.
   The American Psychiatric Association enumerates three criteria that must be met in determining the presence of emotional and behavioral disorders, particularly among adults:
   - *The person experiences significant pain or distress, an inability to work or play, an increased risk of death, or a loss of freedom in important areas of life.*
   - *The source of the problem lies within the person, due to biological factors, learned habits or mental processes, and is not simply a normal response to specific life events such as the death of a loved one.*
   - *The problem is not deliberate reaction to conditions such as poverty, prejudice, government policy or other conflicts with society.*
2. ***Quay's Statistical Classification***
   The statistical analysis of data revealed four clusters of traits and behaviors among children with emotional and behavioral disorders:
   a. *Conduct disorder* is characterized by disobedience, being disruptive, getting into fights, being bossy, and temper tantrums.
   b. *Anxiety withdrawal* sometimes called anxiety disorder is manifested by social withdrawal, anxiety, depression, feelings of inferiority, guilt, shyness, and unhappiness.
   c. *Immaturity* shows in short attention span, extreme passivity, daydreaming, preference for younger playmates, and clumsiness.
   d. *Socialized aggression* is marked by truancy, gang membership, theft, and a feeling of pride and belonging to a delinquent subculture.
3. ***Direct Observation and Measurement***
   a. *Frequency* indicates the rate at which the behaviors occur and how often a particular behavior is performed.
   b. *Duration* is the measure of the length and amount of time a child exhibits the disorder behaviors.
   c. *Topography* refers to the physical shape or form of behavior. The topography displayed by children with emotional and behavioral disorders deviate from the normal.
   d. *Magnitude* refers to the intensity of the displayed behavior. The magnitude of one's aggressive behavior may have a high magnitude or intensity.

e. *Stimulus control* refers to the inability to select an appropriate response to a stimulus. The child has difficulty in selecting the correct behavior in a social stimulus.

4. ***Degree of Severity***

Studies conducted by Olson, Algozzini and Schmid (1980) indicate that emotional and behavioral disorders can be mild and severe. The children who respond positively to therapy and intervention have a mild level or degree of emotional and behavioral disorders. They can attend regular classes and work successfully with the regular and special education teacher and the guidance counselor.

Etiological Factors and Causes of Emotional and Behavioral Disorders

Biological Factors

Authorities believed that all children are born with a biologically determined temperament. The inborn temperament may not be directly causing a behavioral problem to occur but may predispose the child to behavior disturbances. Certain events that easy-going children can handle may be problematic to other children with a difficult temperament. However, studies show that even when a biological impairment exists, there are no proofs to link the physiological abnormality to the occurrence of emotional and behavioral disorders.

Environmental Factors

Home and Family Influence

Studies present evidence on the correlation between parent-child interaction patterns and the development of positive and normal behavioral characteristics in the child. Frequent parental involvement in providing for the child's physical and psychological needs is a significant factor in developing a healthy self-concept.

School Experiences

Research data shows that classroom experiences can maintain and strengthen behavior problems even though the teachers try to control the situation. The child's behavior pattern learned in school is a composite of behavior and attitudes learned at home that interact with his or her experiences with different teachers and classmates.

Characteristics of Children and Youth with Emotional and Behavioral Disorders

Intelligence, Intellectual Characteristics, and Academic Achievement

The following general outcomes describe the intellectual and academic abilities of children (Heward, 2003).

- *Two-thirds could not pass competency examinations for their grade level.*
- *They have the lowest grade point average of any group of students with disabilities.*
- *Forty-four percent failed one or more courses in their most recent school year.*
- *They have a higher absenteeism rate than any other disability category, missing average of 18 days of school per year.*
- *Forty-eight percent drop out of high school, compared to 30% of all students with disabilities and 24% of all high school students.*
- *Over 50% are not employed within 2 years of exiting school.*

Social Skills and Interpersonal Relationships

Studies confirm the observation that students with emotional and behavioral disorders often experience great difficulty in developing and maintaining interpersonal relationships as early as during early childhood. The problems in acquiring social skills and in maintaining healthy interpersonal relationships persist through the adolescence period and adulthood. In secondary schools, research findings show that children tend to have low empathy for others as evidenced by their *"I don't care"* attitude towards people they inflict in pain.

Antisocial Behavior

These children manifest consistent and frequent disordered patterns of behavior that violate the rules and regulations at home, the laws of the community and the country. They show their disdain for society and its norms by engaging in activities that goa against others and property. In the classroom where students are expected to follow certain standards, these children maintain an out-of-seat behavior, do not complete schoolwork, run around, hit and pick up fights, disturb their classmates, ignore, talk back to argue with the teachers and school authorities, complain excessively and distort the truth.

Oppositional Defiant Disorder (ODD)

Students or individuals with oppositional defiant disorder consistently go against, oppose, defy, and show hostility towards authority figures. The symptoms are (APA, 1994):

- *Often loses one's temper*
- *Often argues with adult's request or rules*
- *Often actively defies or refuses to comply with adult's request or rules*
- *Often deliberately annoys people*
- *Often blame others for one's mistakes or misbehavior*
- *Often touchy or easily annoyed by others*
- *Often angry and resentful*
- *Often spiteful and vindictive*

Externalizing and Internalizing Behavior Disorders

Some children with emotional and behavioral disorders display externalizing behavioral disorders that violate the rules and norms of society and annoy and disturb other people. Some common examples are: out-of-seat behavior, making unnecessary noise, truancy, constant talking to self and others, disobedience, inattention, persistent lying, constant blaming of others. On the other hand, too little social interaction of children with internalizing behavioral disorders creates a serious impediment to their development. They manifest withdrawn behavior, lack social skills, often daydream, tend to be fearful of things and events without reason and may experience serious bouts of depression. Internalizing behavioral disorders involve mental or emotional conflicts that may go unnoticed.

Aggressive and Violent Behavior
Aggression refers to acts that are abusive that severely interfere with the activities of other people or objects and events in the environment. Examples of the milder forms of aggression are teasing, clowning around, tattling, and bullying. Severe aggression includes threat of physical harm, physical attack, destruction of property and cruelty.

Delinquency
Delinquency is a legal term that refers to the criminal offenses committed by an adolescent. Studies show that a pattern of antisocial behavior early in a child's life is a strong predictor of delinquency in adolescence. Criminal careers start at an early age, usually by age 12. Examples of delinquent acts and the crimes they can lead to include:

Juvenile Offenses	Crime
1. breaking in and destroying private property, attempted burglary, stealing, shoplifting	1. robbery
2. brutality – beating up a person until he or she is black and blue, burning a house or a person, shooting a person	2. attempted homicide, murder
3. lascivious acts, touching the private parts of a person, attempted rape especially of children, those with disabilities	3. rape
4. early smoking and drinking, experimenting habituation to prohibited drugs	4. committing crimes under the influence of liquor, drugs, drug dependency, drug pushing
5. carrying a knife, ice pick	5. carrying deadly weapons
6. disorderly conduct	6. shooting incidents, murder

Assessment Procedures
Prereferral intervention consists of teacher nomination, parent, and peer nomination using checklists of behavior and learning characteristics at home, in school and other typical environments.

A battery of assessment materials is used in the multifactored evaluation that covers achievement and intelligence tests, social development and personality tests.

Bautista (2003) developed a Behavior Checklist for the Identification of Pupils with Hyperactivity from Grades I to IV. The 45 items measure the extent of hyperactive behavior based on the time and frequency rates of temper outbursts, restlessness, shifting from one task to another, bullying and teasing, fidgeting, oversensitivity, and other related behavior.

Rigonan (2002) developed an Aggression Inventory Scale for Adolescents that measures hostility, disobedience, destructiveness, antisocial tendencies, and dominance.

Ibañez (2003) developed the Deviant Behavior Tendencies Scale that determines the range of deviant behavior as manifested in acts such as defacing school property, assaulting, or abusing students and school authorities, wearing, or displaying unacceptable attire and grooming, engaging in activities that interfere with academic performance, achievements, and violate legal norms and policies.

Educational Approaches
Applied Behavior Analysis

The regular teacher and the special education teacher work collaboratively in developing an individualized education plan or IEP. The aim is to decrease the undesirable and maladaptive behavior and increase the occurrence of desirable behavior. The behavioral theory and model of personality development is applied. The theory assumes that the behavior problems have been learned from his or her history of interactions with the environment. Applied behavior analysis strategies are employed to help the child learn new, appropriate responses, and eliminate the inappropriate ones.

Teaching Social Skills

Stephens (1992) has developed a curriculum that covers 132 specific social skills for school-aged children grouped into 30 subcategories under four major areas:

1. *Self-related behaviors: accepting consequences, ethical behavior, expressing feelings, positive attitude toward self*
2. *Task-related behaviors: attending behavior, following directions, performing before others, quality of work*
3. *Environmental behaviors: care for the environment, dealing with emergencies, lunchroom behavior*
4. *Interpersonal behaviors: accepting authority, gaining attention, helping others, making conversations*

Alternative Responses

Knapczyk developed the alternative responses strategy in training four students with behavior problems to handle or defuse provocative incidents. Instructions consisted of individualized videotape modeling and behavior rehearsals. Two male students who were leader in the school served as actors. One played the role of the subject, simulating his usual reactions to provoking situations and demonstrating appropriate alternative responses. The other actor acted out the usual reactions of classmates. After watching the videotapes, the subject students discussed the circumstances of the incidents with their special education teacher and practiced specific alternative responses.

Teaching Self-management Skills

The teaching self-management skills aims to enable students with behavior problems to have some control over their own behavior problems and over their environment. The special education teacher plays the role of the external control agent who teaches appropriate behaviors in the resource room that the student needs to apply in appropriate settings. The teacher cannot be with the student at home, in the classroom or in places in the community and other settings where the student needs to exhibit the learned behaviors. But the one person who is always with the student is his own self. The students learn to observe and record his own behavior in different settings. The records are analyzed together with the teacher so that the student sees for himself the negative effects of his acting out behaviors.

Intervention Procedures that Minimize Behavior Problems

Ecological Intervention (Cullata, et al. 2003)

It is built on the principle that behavior problems exist within the child's environment where a constant global interaction between the child and the environment occurs. The point of encounter between the child and the people or events in the environment are identified. Then, the cultural source of the problem is traced in terms of the people, the cultural practices, and other influences in the community. Finally, an intervention procedure that focuses in the person and the environment is developed and applied to the problematic situation.

Positive Reinforcement

It is a universally accepted intervention designed to increase the display of desirable behavior and decrease or reduce the opportunities for negatively viewed behavior to occur through a system of rewards. External reinforcers are immediately applied when the desirable or preferred behavior is manifested. When the preferred behavior is exhibited regularly, the reinforcers are removed following a systematic schedule. On the other hand, negative reinforcement involves the removal of a negative stimulus contingent upon the desired behavior. Thus, a child can have recess when he or she finishes the seatwork. Extinction is useful in reducing the number, intensity, or duration of an undesirable behavior.

Rule Setting

It is an easy and effective way to manage behavior in the classroom. A few simple, realistic, and easy-to-follow rules are set together with the consequences if they are followed or violated. Positive reinforcement is applied when the rules are followed. When a rule is broken, the teacher uses body language to send the message by stopping the lesson, eye contact, and other ways of showing displeasure.

Pacing the lesson and using a variety of activities

It is simple yet effective ways of managing behavior. Some activities use games, humor, proximity control and letting others follow the examples. The teacher can model the desired behavior without verbal cues. Other educational strategies found to be effective are cognitive strategy and cognitive model. In cognitive strategy, self-monitoring, self-instruction, and self-control strategies are utilized. The goal is to help the students develop self-awareness and self-direction by using positive reinforcement for social development and improved academic performance. The students record his or her own behavior or academic scores. Sometimes the teacher does the recording, and they compare notes of their ratings.

Application:

Cite the definitions of emotional and behavioral disorders according to:

a. Eli Bower (1957)
b. Council for Children with Behavioral Disorders (CCBD, 1989)

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems overdone.	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

Adapted from: http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf

Summary of the Lesson:

The variations in human behavior are influenced by the basic determinants of personality development: (1) the person's genetic background or heredity, (2) environmental factors, and (3) the general patterning of development.

Personality development is influenced by a person's socialization experiences. He or she interacts with various modes of behavior patterns that cover a wide range of both positive and negative experiences. His or her behavior patterns reflect much of his or her personality traits which in turn generate either positive or negative responses from the social groups. The social relations vary from acceptance to rejection, inclusion to exclusion and similar reactions.

Ideally, social roles should be clear and comfortable for the person. However, social roles can be conflicting, unclear, and not understood so that the end results are misunderstanding, quarrel, discomfort, negative feelings, and unhealthy attitudes. In such cases, a person's personality development and adjustment patterns may be impaired.

Assessment:

I. Identification. Write the name of person or words that is being described in each statement.

_____________________1. It refers to a person's behavior patterns that have desirable consequences and foster his or her well-being and ultimately that of the group.
_____________________2. It is the public law on special education in America that defines serious emotional disturbance.
_____________________3. This refers to the time that elapses between the opportunity to respond and the beginning of the behavior.
_____________________4. An act that are abusive and severely interfere with the activities of other people or objects and events in the environment.
_____________________5. This consists of teacher nomination, parent, and peer nomination using checklists of behavior and learning characteristics at home, in school and other typical environments.

II. True or False. Write True if the statement is correct and False if otherwise. Write the answer before each number.
_____________________1. Behavior problems, emotional and behavioral disorders are manifestations of maladaptive behavior.
_____________________2. Measuring and interpreting disorder behavior across time and setting is easy, exact, and very cheap endeavor.
_____________________3. The DSM–IV is an elaborate clinical classification system consisting of 150 separate diagnostic categories or labels to identify the various types of disordered behavior.
_____________________4. Studies confirm the observation that students with emotional and behavioral disorders often experience great difficulty in developing and maintaining interpersonal relationships as early as during early childhood.
_____________________5. Criminal careers start at an early age, usually by age 18.

Enrichment Activity:

Write your own definition of emotional and behavioral disorders.

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.

		overdone.		
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

Adapted from: http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf

Suggested Links (Optional):

- *https://www.lynchburg.edu/wp-content/uploads/volume-5-2010-11/JamesR-Emotional-Behavioral-Disorders.pdf*

References/Attributions:

- Inciong, Teresita G., Quijano, Yolanda S. and Capulong, Yolanda T. Introduction to Special Education.
  Manila, Philippines: Rex Bookstore, 2007.

Lesson 2: Hearing Impairment

Lesson Objectives:

1. define the terms hearing impairment, hearing loss, deaf and hard of hearing;
2. name the parts of the hearing mechanism and their functions;
3. describe the normal process of hearing audition;
4. explain the role of hearing and listening in language development;
5. enumerate and describe the causes and classifications of hearing loss;
6. explain the effects of hearing impairment on intellectual, social and emotional development;
7. enumerate and describe the formal and informal hearing tests;
8. enumerate and describe the assessment procedures in determining the cognitive ability, communication skills, and socio-behavioral traits of students with hearing impairment;
9. enumerate and describe the types of educational programs, philosophical approaches and instructional strategies for students with hearing impairment;
10. cite the importance of support services in the education of students with hearing loss; and
11. appreciate the abilities of persons who are deaf and hard of hearing.

Getting Started:

Suhana's Story
(The following vignette illustrates the early intervention and special education significantly reduces the effects of hard of hearing in the life of the young girl.)

Suhana Alam is a member of CDC's Early Hearing Detection and Intervention (EHDI) team, serving as an epidemiologist, Division of Human Development and Disability (DHDD), National Center on Birth Defects and Developmental Disabilities (NCBDDD). Suhana identifies as a person who is hard of hearing since birth, and also has quite the story in terms of her own hearing.

The story starts with Suhana's sister, Shahrine Khaled, who is older by 18 months. While their mother was pregnant with Suhana, their uncle came to town for a visit. During the visit, their uncle was quick to notice that Shahrine did not seem to be talking at an age-appropriate level, respond when called upon, and would routinely turn up the volume on television and radio. Shahrine's parents thought that her speech development and behavior were normal for a toddler, but thanks to the uncle expressing his concerns, the family soon took action. A hearing test confirmed that Shahrine was hard of hearing. When Suhana was born, she received a hearing screening and was discovered to be hard of hearing, as well.

Had it not been for the concerns raised by the children's uncle, not only would Shahrine's hearing loss have possibly gone on longer without being detected, but Suhana would most likely not have had a hearing screening at birth. Due to the early intervention and collaborative efforts from a team of physicians, speech therapists, counselors, and teachers, Suhana and Shahrine's parents were equipped with the knowledge they needed to make sure both of their children could reach their full potential in life.

Suhana credits her parents for her own successes, saying that she couldn't have made it as far as she has without their support and patience. And indeed, she has made it far. In April 2015, Suhana achieved what she feels is her greatest accomplishment to date: being hired by the EHDI team as an epidemiologist with full time employee (FTE) status, after being an ORISE fellow since 2012.

Discussion:

Students with hearing impairment are either deaf or hard of hearing. Students who are deaf do not have sufficient residual hearing to understand speech without special instruction and training. Students on the other hand, students who are hard of hearing have enough residual hearing to understand speech and learn in a regular class without much difficulty.

Hearing impairment is not simply an inability to hear or to communicate through speech. The most devastating effect of deafness is the deprivation of language. A hearing person acquires the complex linguistic system of his or her culture as part of normal growth and development in a spontaneous effortless and natural manner. Deafness, on the other hand deprives the person of the normal use of hearing mechanism. He or she does not acquire the listening skills that provide the base for the development of speaking, reading, writing, and other communication competencies. The condition brings about corollary problems in cognitive development, emotional adjustment, difficulties in socialization and anxiety in daily experiences that only a person with hearing impairment can describe.

Helen Keller who was deaf and blind, described the problems of deafness as "deeper and more complex… a much worse misfortune for it means the loss of the most vital stimulus, the sound of the human voice that brings language, sets thoughts astir, and keeps us in the company of man".

Definition of Hearing Impairment or Disability, Deaf and hard of Hearing

Hearing impairment or disability refers to the reduced function or loss of the normal function of the hearing mechanism. The impairment or disability limits the person's sensitivity to tasks like listening, understanding speech, and speaking in the same way those persons with normal hearing do.

According to the age of onset, hearing impairment can be a congenital when the condition is present at birth or adventitious when it is acquired after birth or later. The time when the hearing impairment occurs in terms of the normal development of spoken language at the age of two or thereabout is another classification of deafness. When the condition occurs before the child learns to talk, deafness is prelingual. Deafness is post lingual when it is acquired after child has learned speech usually at the age of two.

A person who is deaf cannot use hearing to listen, understand speech and communicate orally without special adaptations mainly in the visual mode. While a hearing aid amplifies the sounds by increasing the volume to make the sounds louder, a person who is deaf cannot understand speech through the ears alone. He or she may be able to perceive some sounds but his or her sense of hearing is not enough or nonfunctional for the ordinary purposes in life. Speech is accompanied by visually perceived actions like gestures, signs, and facial expression.

A person who is hard of hearing has a significant loss of hearing sensitivity but he or she can hear sounds, respond to speech and other auditory stimuli with or without the use of hearing aid. He or she is more like a hearing person than one who is deaf because both of them use audition or listening to auditory stimuli in the environment, unlike a deaf person who relies more on visual stimuli.

The concept of hearing impairment is often misunderstood. Hearing impairment itself is mistakenly attributed to sub-average intellectual capacity, speech defect, inattention and other learning problems. While deafness adversely affects educational performance because of the difficulty in processing linguistic information through hearing with or without amplification, it is not the same as mental retardation, speech ang language disorders or learning disabilities.

Hearing impairment brings about a diverse group of individuals with special needs. The etiology of hearing impairment, the degrees of hearing loss, and other factors affect normal growth and development in general and speech and language in particular. There is a concomitant effect on social adjustment as communication becomes more difficult due to the decrease in hearing sensitivity.

The following table shows the intensity of sound as heard by persons who have hearing impairments. The severity or the degree of the hearing loss corresponds to loses in decibels that result to learning difficulties.

Severity of Hearing Loss and Resulting Impairment

Degree of Hearing Loss Normal	**Decibel Loss** 0-20dB	**Resulting Impairments**
Slight	27-40dB	Faint sounds and distant conversations are difficult to hear. With a hearing aid, the student can attend regular school.
Mild	41-55dB	As much as 50 percent of classroom conversations are missed. Limited vocabulary and speech difficulties may result.
Moderate	56-70dB	Loud conversations can be heard. Defective speech, language difficulties and limited vocabulary may result.
Severe	71-90dB	Hearing is limited to a radius of one foot, enough to discriminate loud sounds. Defective speech and language and severe difficulty in hearing consonant sounds may result.
Profound	91- and above	Sounds and tones cannot be perceived. Vision becomes the primary sense of communication. Speech and language are likely to deteriorate.

Incidence and Prevalence

In the Philippines, the conservative estimate is that 2%of the population has hearing impairment and the number may increase if children below school age and persons who lose hearing sensitivity due to old age are included.

In the united states at least in every 22 newly born infants has some degree of hearing impairment. At least 3 in 1000 infants have a severe or profound hearing impairment. In its latest report, the US Department of education claimed that children with hearing impairment constitute 1.3% of pupils provided with special education services and 11% of the total age population.

Etiology of Hearing Impairment

Hearing impairments are attributed to genetic and heredity factors, infections, environmental and traumatic factors. Some hearing impairments have unknown causes.

- **Genetic and hereditary types of deafness** occur in one out of thousand live births. Causes are hereditary and chromosomal abnormalities.
- **Infections** such as maternal rubella, cytomegalovirus, hepatitis B virus, syphilis, mumps, and otitis media may occur during pregnancy or afterbirth.
- **Adventitious hearing loss** can be attributed to environmental factors such as excessive and constant exposure to very loud noises. Drugs and medication that can turn toxic when administered to the mother or to the child at inappropriate times and circumstances. Traumatic factors can cause hearing impairment at birth. Low birth weight, difficult and prolonged labor can traumatize the hearing mechanism and cause hearing loss and

permanent damage to the ear. Skull fractures due to accidents, as well as pressure changes may damage the ear. The more specific causes of conductive hearing loss are otitis media (middle ear infection), excessive earwax (impacted serumen), and otosclerosis (a spongy-boney growth around the stirrup which impedes its movement). Sensorineural hearing loss results from damage to the cochlea or auditory nerve. Other causes are viral diseases, RH incompatibility, hereditary factors, exposure to noise, aging and ototoxic medication.

Characteristics of Persons with Hearing Impairment

Individuals with hearing impairment compose a widely diverse group of persons. Since the major effect of deafness is in language development, concomitant issues occur on intellectual and social development, speech and language development that are closely connected to educational concerns.

Deafness is described as an invisible handicapping condition because there are only a few physical and observable manifestations to indicate its presence such as the absence of the outer ear, closed ear canal and fluid discharge from the ear.

Some of the observable behavioral and learning characteristics of a child with hearing impairment are as follows:

- Cups hand behind ear, cocks' ear/tilts head at an angle to catch sounds
- Has strained or blank facial expression when listening or talked to
- Pays attention to vibration and vibrating objects
- Moves closer to speaker, watches face especially the mouth and the lips of the speaker when talked to
- Less responsive to noise, voice, music and other sources of sounds
- Uses more natural gestures, signs and movements to express self
- Shows marked imitativeness at work and play
- Often fails to respond to oral questions
- Often asks for repetition of questions and statements
- Often unable to follow oral directions and instructions
- Has difficulty in associating concrete with abstract ideas
- Has poor general learning performance

As mentioned earlier, the primary effect of hearing impairment is on the development of speech and the acquisition of language skills. In listening, speech and communication, reading and writing. Speech is usually labored, unintelligible, unpleasant and difficult to understand. Vocabulary is limited with problems in syntax. Speech sounds telegraphic and has poor rhythm. There are problems in articulation such as omission, addition, and substitution of letters and sounds or distortion of the words. Poor reading ability results to difficulties in learning the other school subjects.

Studies show that the educational achievement of students with hearing impairment is three to four years below the age-appropriate levels of their hearing peers. Appropriate special education curriculum, instructional strategies and support services help reduce the lags in acquiring the skills in the basic elementary curriculum.

Those with slight or mild hearing losses have a good amount residual hearing to, learn the skills in the curriculum. Deaf persons complain that hearing aids amplify sounds but do not necessarily make them clear and intelligible.

Socio-emotional development follows the same stages among children with hearing impairment. However, lack of communication skills hamper socialization and interaction with hearing persons. The situation forces persons with hearing impairment to stay in each other's company giving rise to a "culture of deafness." As the children mature the demand for the use of speech grows and catching up becomes almost impossible, isolating them further from the mainstream of society. The inclusion of socialization activities in the curriculum and mainstreaming deaf children in regular classes increase the opportunities for socio-emotional development. Likewise, hearing students learn to accept those with hearing impairment as their peers.

Identification and Assessment of Children with Hearing Impairment

Early identification of a hearing impairment increases the chances for the child to receive early treatment and special education intervention. The assessment program includes audiological evaluation, test of mental ability, and test of communication ability.

1. Audiological Evaluation

Audiology is the science of testing and evaluating hearing ability to detect and describe hearing impairments.

Audiological evaluation is done by an audiologist through the uses of sophisticated instruments and techniques. The **audiometer** is an electronic devise that generates sounds at different levels of intensity and frequency. The purpose of audiological evaluation is to determine frequencies of sounds that a particular person hears.

Pure tone audiometry utilizes pure tones in air and bone conduction tests which yield quantitative as well as qualitative description of a child's hearing loss.

Another audiometric test is speech audiometry which uses speech instead of pure tones. Here, the person's detection of speech at the minimum audible level is measured. The understanding of speech sound and the ability to discriminate different speech sounds under sufficient loudness are also determined.

There are *alternative audiometric techniques* for hearing evaluation such as *the sound field audiometry, evoked response audiometry, impedance audiometry, play audiometry, operant conditioning audiometry and behavior observation audiometry.*

In the Philippines where formal audiological services are limited, informal tests of hearing are employed. The procedures for some of these hearing tests are:

Informal Hearing Tests

a. Whisper test

Sit the child comfortably. Ask him or her to stick the tip of the forefinger in one ear.
The tester sits behind the child where the uncovered ear is. After a deep breathe whisper some familiar words that contain high pitch and low pitch tones right behind the unblock ear. The child must be able to repeat the words correctly.

b. Conversational live voice test

Keeping the same position but facing the child, ask him or her to repeat words that contain high and low pitch consonants. Start with a whisper and increase the intensity up to 20dB moving away from child little by little. If the child hears at a distance of 3 to 6 meters, hearing is normal. If the child can repeat the words but speech is unclear, he or she might be hard of hearing.

c. Ball pen click test

Use a retractable ball pen and place it one inch away from the ear. While the other ear is blocked by a finger, press the button of the ball pen down and release it. Do it only once. The child indicates that he or she hears the click by either raising one hand or acknowledging it with a yes or a nod.

1. Cognitive Assessment

The assessment tools that measure intellectual capacity of children with hearing impairment do not rely primarily on verbal abilities. In the United States, the Hiskey- test of learning Aptitude, the Wechsler Intelligence Scale for children (WISC) and the Stanford Achievement Test (SAT) are widely used because of the nonverbal performance subtests.

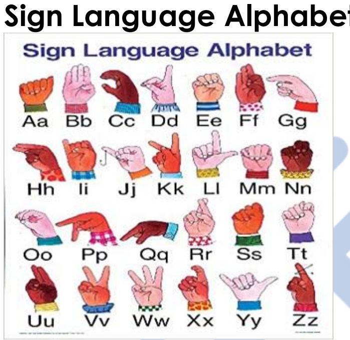
2. Assessment of Communication Abilities

Assessment of speech and language abilities includes an analysis of the development of the form, content and use of language. Articulation, pitch, frequency, and quality of voice are examined.

3. Social and Behavioral Assessment

Hearing impairment brings about significant effects on social-emotional and personality development as a result of the restrictions in interactive experiences and communication activities maladjustment with their age group. Linguistic difficulties often times show in low self-concept and social- emotional maladjustment.

Sign Language Alphabet



The degree and classification of hearing loss are important factors in declining the most appropriate special education program for children with hearing impairment.
There are other considerations that need attention in the education of students with hearing impairment. Some of them are educational environment, mode of communication and support services.

Educational Placement

In the Philippines students with hearing impairment like other students with disabilities are mainstreamed in regular classes either on full-time or part-time basis. A special education teacher assists the regular teacher in seeing to it that students receive as much instruction as their hearing classmates. Some special education programs employ an interpreter in the regular class who translates the verbal activities into signs and gestures to enable the student to follow the lesson. The special education teacher gives special instruction in the resource room or in the Special Education

Center on oral or total communication, manual communication that includes finger spelling and different signed systems, auditory-verbal training and cued speech.

Support Services

Communication accessibility is provided by sign language and oral interpreters inside and outside the classrooms. Computer-aided instruction (CAI) reinforces the knowledge and skills learned in the different subject areas. The special learning areas like speech, auditory training and language are taught effectively through computer-aided instruction.

Some TV programs and video utilize closed captioning services in sign language and make the programs accessible to individuals with hearing impairment. Telephone services have telecommunication devises for the deaf where visual display of communication going on is available in print and electronic display. Cellular phones with text messaging features are very useful to individuals with hearing impairment.

Suggestions for Teaching Students with Hearing Impairment in a Regular Class

1. Promote the acceptance of the student with hearing impairment in the regular class.
    - Welcome the student to your class.
    - Explain the student's condition to the entire class. Emphasize that he or she can learn together with hearing students.
    - Make modifications in teaching as natural as possible.
    - Accept the student as an individual with abilities and limitations.
    - Discuss the student's condition with him or her
2. Be sure that prescribed hearing aids and other amplification devices are used.
    - Understand and explain to the class that the hearing aid makes sound louder but not necessarily clearer.
    - See to it that the special education teacher checks the student's hearing aid or other devices and that they are working properly.
    - Encourage the student to take care of his or her hearing aid and to tell the teacher when it is not functioning properly.
    - Be sure your student has a spare battery at school.
    - Tell the special education teacher or the parents if the student's hearing aid is not working properly.
3. Provide preferential seating.
    - Sit the student near the spot where you typically stay when teaching.
    - Sit the student where he or she can easily watch your face without straining to look up.
    - Sit the student away from sources of noise.
    - Sit the student where light is on your face and not in the student's eyes.
    - Sit the student so he or she can use the better ear.
    - Allow the student to transfer to other seats when necessary.
4. Increase visual information
    - Remember that your student reads your lips and must see your face in to do so.

- Try to stay in one place while talking to the class so the student does not have to lipread a "moving target".
- Avoid talking when your back is turned to the class such as when you are writing on the chalkboard.
- Avoid covering your mouth or face when talking.
- When reading in front of the class, be sure that the student can lipread you.
- Avoid standing in front of windows where the glare will make it difficult to see your face.
- Use visual aids, such as pictures and illustrations whenever possible.
- Demonstrate what you want to student to understand whenever possible.
- Use the chalkboard as much as possible.

5. Minimize classroom noise.
    - Set the student away from noisy parts of the classroom.
    - Wait for the class to be quiet before talking to the students.
6. Modify teaching procedures.
    - Be sure that student is watching and listening when you are talking.
    - Be sure that student understands what is said by asking him or her to repeat information or answer questions.
    - Rephrase rather than repeat questions and instructions whenever necessary.
    - Write key words, new words, and other needed information on the chalkboard.
    - Repeat or rephrase things said by other students during classroom discussion.
    - Introduce new vocabulary words to the student in advance.
    - Ask the student to repeat if you cannot understand him or her.
    - Assign a student as "buddy" to alert the deaf student to listen and to be sure that he or she understand the lesson correctly.
7. Have realistic expectations.
    - Remember that the student cannot understand and grasp everything all of the time, no matter how hard he or she tries.
    - Be patient when the student asks for repetition.
    - Give the student a break from listening when he or she knows signs of fatigue.
    - Expect the student to follow classroom routine.
    - Expect the student to abide by all the school rules such as attendance, proper behavior, homework, and dependability as other students are required to do.
    - Be alert for fluctuations of hearing. Report any observation to the special education teacher.

Educational Approaches Used when Working with students with Hearing Impairment

	Basic Position	**Objective**	**Method of Communication**
Bilingual-Bicultural	Considers American Sign Language (ASL) to be the natural language of the Deaf culture and urges	To provide a foundation in the use of ASL with its unique vocabulary and syntax	ASL (American Sign Language)

	recognition of ASL as the primary language choice with English considered a second language	rules; ESL instruction provided for English vocabulary and syntax rules	
Total Communication	Supports the belief that simultaneous use of multiple communication techniques enhances an individual's ability to communicate, comprehend and learn.	To provide a multifaceted approach of communication to facilitate whichever method(s) works best for each individual	Combination of sign language (accepts the use of any of the sign language systems), fingerspelling, and speechreading.
Auditory-Aural	Supports the belief that children with hearing impairment can develop listening/receptive language and oral language expression (English) skills; emphasizes use of residual hearing (the level of hearing an individual possesses), amplification (hearing aids, auditory training), and speech/language training	To facilitate the development of spoken (oral) English	Spoken (oral) English
Source: Garguilo, R., Special Education in Contemporary Society, p,430			

Application:
Find out how much you learned about hearing impairment by answering the question below.

1. What is hearing impairment? Why is early identification of hearing loss important?

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.

Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems overdone.	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

Adapted from: http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf

Summary of the Lesson:

The term "hearing impairment" refers to functional hearing loss that ranges from mild to profound. Hearing loss is a partial or total inability to hear. Hearing loss may be present at birth or acquired at any time afterwards. Often, people who have no functional hearing refer to themselves as "DEAF." Those with milder hearing loss refer to themselves as "HARD OF HEARING." Hearing loss may occur in one or both ears. In children, hearing problems can affect the ability to acquire spoken language, and in adults it can create difficulties with social interaction and at work. Hearing loss can be temporary or permanent. Deaf people usually have little to no hearing.

Hearing loss may be caused by a number of factors, including: genetics, ageing, exposure to noise, some infections, birth complications, trauma to the ear, and certain medications or toxins.

Accommodations for students with hearing impairments can be classified as VISUAL and AURAL. Visual accommodations rely on a person's sight; aural accommodations rely on a person's hearing abilities. Examples of visual accommodations include sign language interpreters, lip reading, and captioning. Examples of aural accommodations include amplification devices such as hearing aids.

Assessment:

Myths and Facts About Hearing Impairment

Direction: Write **F** if the statement is a Fact; otherwise write **M**.

1. Hearing impairment is classified according to the part of the ear where the defect is, the age at onset, the stage of language development at the time of onset, and the degree of hearing impairment.
2. It is best to pick only one language for Deaf or Hard of Hearing children, such as English or American Sign Language.
3. Students who are Deaf or Hard of Hearing and wear hearing aids in the classroom can hear everything just fine.
4. Hearing loss affects language development that in turn leads to delays in cognitive, social and emotional development.

5. Special education covers a wide range of educational approaches and strategies that focus on the development of speech, language and communication skills.
6. All individuals who are Deaf or Hard of Hearing can read lips – I just have to look at them directly to make sure they can understand.
7. Hearing loss does not affect new born babies.
8. The two main categories of hearing impairment are deafness and hard of hearing.
9. Only very loud noises causes hearing loss.
10. Hearing loss only affects old people and is just a sign of aging.

Enrichment Activity:

What are the advantages of having intact sensory modalities especially audition/hearing? What do you do to preserve your audition?

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
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Adapted from: http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf

References/Attributions:

▪ Inciong, Teresita G., Quijano, Yolanda S. and Capulong, Yolanda T. Introduction to Special Education. Manila, Philippines: Rex Bookstore, 2007.

Lesson 3: Visual Impairment

Lesson Objectives:

1. describe the anatomy and physiology of the human eye and how the process of vision takes place;
2. define legal and educational blindness;
3. differentiate low vision from blindness;
4. enumerate and describe the types and causes of the problems of vision;
5. enumerate and describe the advances in technology for blind persons; and
6. describe the educational provisions for students with visual disabilities.

Getting Started:

XAVI

"I DID IT"

(This success story about a little who is visually impaired illustrates that a disability need not be a stumbling block in acquiring an education.)

Xavier spontaneously exclaimed those three words as he walked away with his diploma from the Blind Children's Center. For the eight-year-old, the moment was the culmination of a long and often arduous journey. Born prematurely with multiple disabilities including Cerebral Palsy and Cortical Visual Impairment, Xavier's early school years were unlike most, even at the Blind Children's Center.

Before graduation day, he had already battled through several major surgeries including hip adductor and hamstring lengthening. He had worn leg braces for six to eight hours a day at school since the time he was crawling, and just last year he endured another surgery to improve the mobility in his ankle. Despite all those challenges, he may have been the student who spent the most time at school smiling. He quickly earned the nickname "Happy Xavi"

"He really is a very happy little boy," his mother Carla said. "And the teachers and specialists at the Center have been so patient and caring with him." The family workers and teachers visited during his hospital stays, brought him his schoolwork, and helped walk her through the medical process. "They provided light through some scary times," she said.

Xavier loves to tell jokes and has an uncanny memory for matching names to voices, including nearly ever staff member at the Blind Children's Center! Although he has significant delays, he enjoys stories and has learned to spell his name by rote.

Most incredibly, after several years using a walker, he took his first steps on his own last year, into his grandmother's arms. "When he was born the doctors said he would never walk," Carla remembered. "I know so much of his growth has to do with the specialists and their adaptive physical education program."

"Xavier's success is a great example of the entire family working together with our multidisciplinary team," said the Center's Director of Education & Family Services, Dr. Fernanda Armenta-Schmitt. "His mother, grandmother and aunt are frequently on campus, benefitting from our family groups and community of support."

Xavier just started at his local public elementary school. His aunt, who also lives with Xavi, said he already likes it very much though he does miss the Blind Children's Center. "Although he is in a class with children of varying abilities at his new school, he was prepared for that thanks to the Center's inclusion model," she said.

"Xavi brings joy to everyone who meets him. When he arrived at the Center six years ago, we never imagined how far he would come. We are very grateful."

Discussion:

Man's capacity to use his or her visual mechanism places him or her on top of all other creatures in the world. There is no doubt that in the list of the basic human senses, vision is number one, followed by audition or hearing, kinesthesia or touch, olfaction or smell and gustation or taste.

Authorities state that although man uses all his senses simultaneously in gathering varied stimuli from the environment nearly eighty percent (80%) of all knowledge and information that man acquires in his or her lifetime are gained through the visual modality. With the use of human intelligence mainly through vision, man has attained superiority over all other species in the world as shown in the tremendous advances in technology through the centuries. A writer poses two questions on how much man values human sight: "Through the centuries, how many have really appreciated God's greatest gift to man? Even in this day of modern scientific miracles and educational opportunities, how many really know what the eye really is and what makes it work? "To which a third question is added: "How much and how well does man take good care of his or her vision?"

Blindness and Low Vision

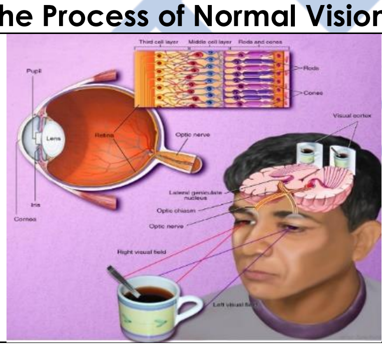
There are two general definitions of blindness. The first is the **legal definition** that is based on measurement of **visual acuity, field of vision and peripheral vision. Visual acuity** is the ability to clearly distinguish forms or discriminate details at a specific distance. Normal visual acuity is measured by reading letters, numbers and other symbols from chart 20 feet away. The Snellen chart is commonly used for this purpose. The sizes of the letters in the chart correspond to the appropriate distances where they can be read with normal vision. Thus, when a person can read the row of letters marked 20/20 correctly, he or she has normal vision. His/her visual acuity is 20/20, that is, he or she can read the letters that normal vision permits to be read 20 feet away when seated 20 feet away from the chart. On the other hand, if a person can read the row of letters marked 20/70 only when seated 20 feet away from the chart, his visual acuity is poor. Normally, the larger letters can be read at a distance of 70 feet. Or, when a person can read the rows of letters marked 20/200 only when seated 20 feet away from the chart, his visual acuity is so poor that it falls within the range of blindness. However, there are people whose visual acuity in one eye is better than normal at 20/10. This means that the better eye can see at 10 feet away what normally can be seen at 20feet away.

The **field of vision refers** to the area that normal eyes cover above, below and on both sides when looking at an object or when gazing straight ahead. The field of normal vision covers approximately a range of 180 degrees. When looking directly at an object, the **central field of vision** is used. **Peripheral vision** covers the outer ranges of the field of vision. A person may have poor central vision but good peripheral vision. **Tunnel vision** results from an extremely restricted field of vision. It is like looking at the objects in the environment through a narrow tube or tunnel. The field of vision can decrease slowly undetected among children and adults over a period of years. A complete eye examination should include both visual acuity and field of vision.

Legal blindness refers to the condition where **visual acuity is 20/200** in the better eye after the best possible correction with glasses or lenses. The **field of vision** whether central or peripheral is limited to an area of 20 degrees or less from the normal 180-degree field. A legally blind person with his or her eye-glasses or contact lenses on can see or 200 feet away. The person experiences difficulties in everyday activities especially in discerning fine details of objects and things in the environment. In the United States, persons who are legally blind are eligible to receive a wide range of benefits from the government. These include special education or vocational rehabilitation services, free mail service and income tax exemption.

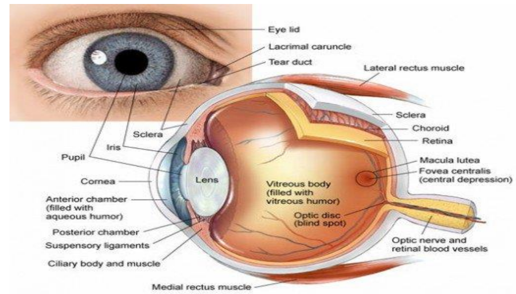
Educational Definition. Not all legally blind persons are totally blind. In total blindness the person is absolutely without sight but may have light and movement perception and travel vision. The degrees of blindness include light perception (person can differentiate between light and dark, day and night), movement perception (person can detect if an object or person is in motion or in still position) and travel vision (field of vision is enough to travel safely in familiar areas). Although classified as blind, the person can still use his or her residual vision.
In special education, children who are blind are differentiated from those who have low vision. Blind children use their sense of touch to read braille and train in orientation and mobility to move around and travel independently. A child with low vision learns to read materials in large print. Corn's definition of low vision (cited in Howard, 2003) emphasizes the functional use of vision. Low vision is a level of vision that with standard correction hinders an individual in the visual planning and execution of tasks, but which permits enhancements of the functional vision through the use of optical or non-optical aids and environmental modifications or techniques.

The Process of Normal Vision



The sense of vision is a complex and intricate physiological system. There are three elements necessary for good vision to take place. These are a pair of healthy, intact, and efficiently functioning eyes with complete parts, well-lighted objects and images and a healthy brain.

Anatomy and Physiology of the Human Eye



The parts of the eye that we see on the face are only a small part of the total mechanism for seeing. The eye is a complex part of the human body that no other organ can equal. There are five physiological or physical systems in vision, namely, (1) the protective structures, (2) the refractive parts, (3) the muscles, (4) the retina and the optic nerve, and (5) the brain where vision takes place.

The protective structures surround the eye to protect it from harm. These are the bony eye socket in the skull and the protruding bones in the cheeks and forehead, the lacrimation system or tear ducts, the eyebrows, eyelids and eyelashes. The **eye socket** which contains the eyeball where most of the parts of the eye are found, is comparable in size to ping-pong ball. It protects the sensitive mechanism for vision from trauma, together with the **bones of the cheeks and forehead.** The **tear ducts or lacrimation system** protect the eye by secreting fluid or tears that clean and keep the eye moist. The **eyelid** moistens and cleans the cornea through blinking. The **eyebrows** and **eyelashes** catch foreign bodies that may enter the eyes.

The refractive structures bend or refract light rays so that the image of the object focuses on the **retina.** These are the cornea, aqueous humour, pupil, iris, lens and vitreous humour. The **cornea** is the curved transparent membrane that protects the sensitive parts of the eye. It is the front window of the eye that starts the process of vision by bending the light rays into patterns or images.

Then the light rays pass through the acqueous humour chamber which is one of two fluid-filled chambers. The **acqueous humour** is a watery liquid that fills the front chamber of the eye to keep the eyeball properly inflated. The **pupil** is a circular hole in the center of the colored **iris.** The pupil constricts or dilates to regulate the amount of light entering the eye. It constricts or closes to a tiny circle in bright light and dilates or opens wide in darkness to allow as much light as possible to enter the eye. The **lens** is a transparent and elastic little envelop of fluid suspended by tiny strong muscles. The lens adjusts its thickness so that both near and far objects can be brought to focus on the retina. The fluid fattens the lens for near vision and makes it flat for far or distant vision. The lens moves at an average of one hundred thousand times a day, which is equivalent to a fifty-mile hike to the leg muscles. Like the acqueous humour chamber, the vitreous humour chamber is filled with fluid. The vitreous humour is a jelly-like substance that has the consistency of an egg white but still clear enough for light to pass through. The vitreous humour fills most of the eye's interior.

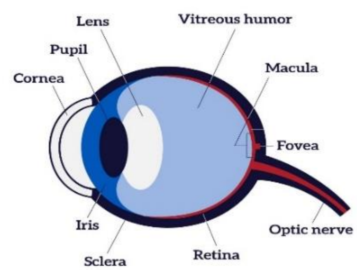
The **muscles** function to coordinate and balance the movements of the eyes. They turn, raise and lower the eyes in response to cranial or brain nerve impulses.

The **retina** is a multi-layered sheet of nerve tissues at the back of the eye. The retina is the last part of the neutral receptor system for vision. It is likened to the film in the camera: for a clear image to be transmitted to the brain, the light rays must come to a precise focus on the central portion of the retina. The retina contains some one hundred thirty-seven million light receptor cells are shaped like rods for black and white vision and seven million are shaped like cones for color perception. The rods are responsible for night vision of for seeing objects and images under conditions of low illumination. All thirty-seven million light sensitive receptors are contained in an area that is less than a square inch. The most sensitive part of the retina is the **macula,** at the center of which lies the **fovea**, the area that is vital to the exact discrimination of the details of an image or object. The **optic nerve** is connected to the retina and conducts visual images to the brain. The optic nerve is capable of transmitting messages from the retina to the brain at a speed of three hundred miles per hour.

Vision takes place in the **occipital lobe of the brain** located at the back of the gead. The occipital lobe is one of the four lobes of the brain that are named for the major skull bones that cover them. Audition or hearing takes place in the temporal lobe found behind the forehead. The parietal lobe at the top of the head toward the rear processes body sensations. The frontal lobe is a part of the cerebral cortex in the cerebrum or forebrain which is the largest part of the brain that governs the highest functions associated with conscious activities and intelligence. The frontal lobe also controls the movement of voluntary muscles.

How Vision Takes Place

All the different parts of your eyes work together to help you see. First, light passes through the cornea (the clear front layer of the eye). The cornea is shaped like a dome and bends light to help the eye focus. Some of this light enters the eye through an opening called the pupil (PYOO-pul). The iris (the colored part of the eye) controls how much light the pupil lets in.



Next, light passes through the lens (a clear inner part of the eye). The lens works together with the cornea to focus light correctly on the retina.

When light hits the retina (a light-sensitive layer of tissue at the back of the eye), special cells called photoreceptors turn the light into electrical signals.
These electrical signals travel from the retina through the optic nerve to the brain. Then the brain turns the signals into the images you see.
Your eyes also need tears to work correctly.

Types and Causes of Problems of Vision

The inability of the eyes to function of the eyes to function efficiently may be traced to (1) errors of refraction; (2) imbalance of the eye muscles; (3) diseases; and (4) trauma or accidents.
1. Errors of Refraction
What Is a Refractive Error?
Refractive error means that the shape of your eye does not bend light correctly, resulting in a blurred image. The main types of refractive errors are myopia (nearsightedness), hyperopia (farsightedness), presbyopia (loss of near vision with age), and astigmatism.
Symptoms

- Blurred vision
- Difficulty reading or seeing up close
- Crossing of the eyes in children (esotropia)

Causes
Overuse of the eyes does not cause or worsen refractive error. The causes of the main types of refractive error are described below:

- **Myopia** (close objects are clear, and distant objects are blurry) Also known as nearsightedness, myopia is usually inherited and often discovered in childhood. Myopia often progresses throughout the teenage years when the body is growing rapidly.
- **Hyperopia** (close objects are more blurry than distant objects) Also known as farsightedness, hyperopia can also be inherited. Children often have hyperopia, which may lessen in adulthood. In mild hyperopia, distance vision is clear while near vision is blurry. In more advanced hyperopia, vision can be blurred at all distances.
- **Presbyopia** (aging of the lens in the eye) After age 40, the lens of the eye becomes more rigid and does not flex as easily. As a result, the eye loses its focusing ability and it becomes more difficult to read at close range. This normal aging process of the lens can also be combined with myopia, hyperopia or astigmatism.
- **Astigmatism** Astigmatism usually occurs when the front surface of the eye, the cornea, has an asymmetric curvature. Normally the cornea is smooth and equally curved in all directions, and light entering the cornea is focused equally on all planes, or in all directions. In astigmatism, the front surface of the cornea is curved more in one direction than in another. This abnormality may result in vision that is much like looking into a distorted, wavy mirror. Usually, astigmatism causes blurred vision at all distances.

Risk Factors

People with high degrees of myopia have a higher risk of retinal detachment which may require surgical repair.

Tests and Diagnosis

A refractive error can be diagnosed by an eye care professional during a routine eye examination. Testing usually consists of asking the patient to read a vision chart while testing an assortment of lenses to maximize a patient's vision. Special imaging or other testing is rarely necessary.

Treatment and Drugs

Refractive disorders are commonly treated using corrective lenses, such as eyeglasses or contact lenses. Refractive surgery (such as LASIK) can also be used to correct some refractive disorders. Presbyopia, in the absence of any other refractive error, can sometimes be treated with over-the-counter reading glasses. There is no way to slow down or reverse presbyopia.

1. Imbalance of the Eye Muscles

When the muscles of both eyes do not work together in a coordinated way, imbalance of the eye muscles occurs. In **strabismus,** different images are cast on each retina resulting to cross-eyedness or squinting. Diplopia or double vision results when the brain cannot fuse the differences in the images cast on the retina into a single image. The condition can be corrected by **prescription lenses, exercises, surgery or a combination of the three.**

Amblyopia occurs when vision is suppressed in one eye and it becomes weak or useless. **Nystagmus** is a condition in which there are rapid involuntary movements of the eyeball that can result to nausea and vomiting and dizziness. In some cases, nystagmus is a sign of brain malfunction or inner ear problem.

Amblyopia vs Strabismus

Very simply, **Strabismus**, the medical term for "crossed-eye", is a problem with eye alignment, in which both eyes do not look at the same place at the same time. **Amblyopia,** the medical term for "lazy-eye", is a problem with visual acuity, or eyesight. Many people make the mistake of saying that a person who has a crossed or turned eye (strabismus) has a "lazy-eye," but lazy-eye (amblyopia) and strabismus are not the same condition.

Both strabismus and amblyopia are treatable conditions by a vision therapy specialist.

Strabismus is the most common cause of amblyopia and amblyopia often occurs along with strabismus. However, amblyopia can occur without strabismus. But, there's more to it than this. Let's take a look at these vision disorders side-by-side.

	Strabismus	**Amblyopia**
Also known as	Crossed-eyes, Squint, wandering eye, deviating eye, walleye	Lazy-Eye
Definition	Strabismus is a condition in which the eyes do not point at the same place at the same time. One or both eyes turn inward (esotropia), outward (exotropia), upward (hypertropia), or downward (hypotropia) either some (intermittent) or all of the time.	Amblyopia is the lack of development of clear vision (acuity) in one or both eyes for reasons other than an eye health problem that cannot be improved with glasses alone.
Types	Types of strabismus are determined by the following: • Which eye turns • Direction of the eye turn • Frequency of the eye turn • Amount of eye turn • Whether the turn is the same in all positions of gaze	All three types of amblyopia result from suppression of vision in one or both eyes. The difference is in the root cause of the suppression. • Refractive amblyopia • Strabismic amblyopia • Deprivation amblyopia
Causes	People can be born with strabismus or it can be acquired later in life. Strabismus can also develop as the result of an accident or other health problem.	Amblyopia begins during infancy and early childhood. The most common causes of amblyopia are:

	Genetics also may play a role: If you or your spouse has strabismus, your children have a greater risk of developing strabismus as well. Strabismus occurs when there are neurological or anatomical problems that interfere with the control and function of the eyes. The problem may originate in the muscles themselves, or in the nerves, or in the vision centers in the brain that control binocular vision. Most cases of strabismus **are not** a result of a muscle problem, but are due to miscommunication between the brain and the eyes. Because the eyes are pointing at different places, the brain has difficulty combining the images from both eyes into a single, 3D image.	• constant strabismus (constant turn of one eye), • anisometropia (different vision/prescriptions in each eye), • and/or blockage of an eye due to trauma, lid droop, etc. Of these, strabismus is the most common cause of amblyopia.
Risk Factors	Risk factors include: • family history of strabismus • prematurity or low birth weight • retinopathy of prematurity • conditions that affect vision, such as cataracts, severe ptosis and corneal scars • muscular abnormalities • neurological abnormalities • amblyopia (or lazy eye) People with parents or siblings who have strabismus are more likely to develop it. People who have a significant amount of uncorrected farsightedness (hyperopia) may develop strabismus because of the additional eye focusing they must do to keep objects clear. People with conditions such as Down syndrome and cerebral palsy or who have suffered a stroke or head injury are at a higher risk for developing strabismus.	Risk factors include: Strabismus and significant refractive errors are risk factors for unilateral amblyopia. Bilateral astigmatism and bilateral hyperopia are risk factors for bilateral amblyopia.
Symptoms	The obvious symptom of strabismus is an observable eye turn.	Typical symptoms include: • Poor depth perception

	Patients with constant strabismus tend to be less symptomatic (but not asymptomatic) when compared to patients with intermittent strabismus. That's because they often suppress the information from the eye that is turning, thus avoiding double vision and other symptoms. Patients with intermittent strabismus may experience more frequent symptoms. These include: • Poor depth perception • Eye strain and/or pain • Headaches • Blurry or double vision • Eye and/or general fatigue Patients with strabismus may report: • difficulty driving, • difficulty reading, • difficulty with sports activities, • feeling clumsier than their peers.	• Difficulty catching and throwing objects • Clumsiness • Squinting or shutting an eye • Head turn or tilt • Eye strain • Fatigue with near work A clue that your child may have amblyopia is if he or she cries or fusses when you cover one eye.
Treatment	Treatment for strabismus may include eyeglasses, prisms, vision therapy, or eye muscle surgery. If detected and treated early, strabismus can often be corrected with excellent results. Vivid Vision is used in the treatment of strabismus.	Treatment for amblyopia (lazy eye) may include a combination of prescription lenses, prisms, vision therapy and eye patching. In vision therapy, patients learn how to use the two eyes together, which helps prevent lazy eye from reoccurring. Vivid Vision is used in the treatment of amblyopia.

Diseases of the Eye

1. Cataract

A cataract is a clouding of the normally clear lens of your eye. For people who have cataracts, seeing through cloudy lenses is a bit like looking through a frosty or fogged-up window. Clouded vision caused by cataracts can make it more difficult to read, drive a car (especially at night) or see the expression on a friend's face.
Most cataracts develop slowly and don't disturb your eyesight early on. But with time, cataracts will eventually interfere with your vision.

At first, stronger lighting and eyeglasses can help you deal with cataracts. But if impaired vision interferes with your usual activities, you might need cataract surgery. Fortunately, cataract surgery is generally a safe, effective procedure.

Symptoms

Signs and symptoms of cataracts include:
Clouded, blurred or dim vision
Increasing difficulty with vision at night

- Sensitivity to light and glare
- Need for brighter light for reading and other activities
- Seeing "halos" around lights
- Frequent changes in eyeglass or contact lens prescription
- Fading or yellowing of colors
- Double vision in a single eye

At first, the cloudiness in your vision caused by a cataract may affect only a small part of the eye's lens and you may be unaware of any vision loss. As the cataract grows larger, it clouds more of your lens and distorts the light passing through the lens. This may lead to more noticeable symptoms.

Causes

Most cataracts develop when aging or injury changes the tissue that makes up your eye's lens. Some inherited genetic disorders that cause other health problems can increase your risk of cataracts. Cataracts can also be caused by other eye conditions, past eye surgery or medical conditions such as diabetes. Long-term use of steroid medications, too, can cause cataracts to develop.

Types of cataracts

Cataract types include:

- **Cataracts affecting the center of the lens (nuclear cataracts).** A nuclear cataract may at first cause more nearsightedness or even a temporary improvement in your reading vision. But with time, the lens gradually turns more densely yellow and further clouds your vision.

As the cataract slowly progresses, the lens may even turn brown. Advanced yellowing or browning of the lens can lead to difficulty distinguishing between shades of color.

- **Cataracts that affect the edges of the lens (cortical cataracts).** A cortical cataract begins as whitish, wedge-shaped opacities or streaks on the outer edge of the lens cortex. As it slowly progresses, the streaks extend to the center and interfere with light passing through the center of the lens.
- **Cataracts that affect the back of the lens (posterior subcapsular cataracts).** A posterior subcapsular cataract starts as a small, opaque area that usually forms near the back of the lens, right in the path of light. A posterior subcapsular cataract often interferes with your reading vision, reduces your vision in bright light, and causes glare or halos around lights at night. These types of cataracts tend to progress faster than other types do.
- **Cataracts you're born with (congenital cataracts).** Some people are born with cataracts or develop them during childhood. These cataracts may be genetic, or associated with an intrauterine infection or trauma.

These cataracts also may be due to certain conditions, such as myotonic dystrophy, galactosemia, neurofibromatosis type 2 or rubella. Congenital cataracts don't always affect vision, but if they do, they're usually removed soon after detection.

Risk factors

Factors that increase your risk of cataracts include:

- Increasing age
- Diabetes
- Excessive exposure to sunlight
- Smoking
- Obesity
- High blood pressure
- Previous eye injury or inflammation
- Previous eye surgery
- Prolonged use of corticosteroid medications
- Drinking excessive amounts of alcohol

Prevention

No studies have proved how to prevent cataracts or slow the progression of cataracts. But doctors think several strategies may be helpful, including:

Have regular eye examinations. Eye examinations can help detect cataracts and other eye problems at their earliest stages. Ask your doctor how often you should have an eye examination.

Quit smoking. Ask your doctor for suggestions about how to stop smoking. Medications, counseling and other strategies are available to help you.

Manage other health problems. Follow your treatment plan if you have diabetes or other medical conditions that can increase your risk of cataracts.

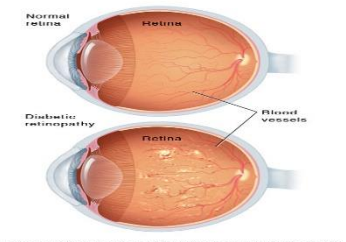
Choose a healthy diet that includes plenty of fruits and vegetables. Adding a variety of colorful fruits and vegetables to your diet ensures that you're getting many vitamins and nutrients. Fruits and vegetables have many antioxidants, which help maintain the health of your eyes.

Studies haven't proved that antioxidants in pill form can prevent cataracts. But a large population study recently showed that a healthy diet rich in vitamins and minerals was associated with a reduced risk of developing cataracts. Fruits and vegetables have many proven health benefits and are a safe way to increase the amount of minerals and vitamins in your diet.

Wear sunglasses. Ultraviolet light from the sun may contribute to the development of cataracts. Wear sunglasses that block ultraviolet B (UVB) rays when you're outdoors.

Reduce alcohol use. Excessive alcohol use can increase the risk of cataracts.

2. Diabetic Retinopathy



Diabetic retinopathy (die-uh-BET-ik ret-ih-NOP-uh-thee) is a diabetes complication that affects eyes. It's caused by damage to the blood vessels of the light-sensitive tissue at the back of the eye (retina).
At first, diabetic retinopathy may cause no symptoms or only mild vision problems. Eventually, it can cause blindness.

The condition can develop in anyone who has type 1 or type 2 diabetes. The longer you have diabetes and the less controlled your blood sugar is, the more likely you are to develop this eye complication.

Symptoms

You might not have symptoms in the early stages of diabetic retinopathy. As the condition progresses, diabetic retinopathy symptoms may include:

- Spots or dark strings floating in your vision (floaters)
- Blurred vision
- Fluctuating vision
- Impaired color vision
- Dark or empty areas in your vision
- Vision loss

Diabetic retinopathy usually affects both eyes.

Causes

Over time, too much sugar in your blood can lead to the blockage of the tiny blood vessels that nourish the retina, cutting off its blood supply. As a result, the eye attempts to grow new blood vessels. But these new blood vessels don't develop properly and can leak easily.

There are two types of diabetic retinopathy:

- **Early diabetic retinopathy.** In this more common form — called non-proliferative diabetic retinopathy (NPDR) — new blood vessels aren't growing (proliferating).

When you have NPDR, the walls of the blood vessels in your retina weaken. Tiny bulges (microaneurysms) protrude from the vessel walls of the smaller vessels, sometimes leaking fluid and blood into the retina. Larger retinal vessels can begin to dilate and become irregular in diameter, as well. NPDR can progress from mild to severe, as more blood vessels become blocked.

Nerve fibers in the retina may begin to swell. Sometimes the central part of the retina (macula) begins to swell (macular edema), a condition that requires treatment.

- **Advanced diabetic retinopathy.** Diabetic retinopathy can progress to this more severe type, known as proliferative diabetic retinopathy. In this type, damaged blood vessels close off, causing the growth of new, abnormal blood vessels in the retina, and can leak into the clear, jelly-like substance that fills the center of your eye (vitreous).

Eventually, scar tissue stimulated by the growth of new blood vessels may cause the retina to detach from the back of your eye. If the new blood vessels interfere with the normal flow of fluid out of the eye, pressure may build up in the eyeball. This can damage the nerve that carries images from your eye to your brain (optic nerve), resulting in glaucoma.

Risk Factors

Anyone who has diabetes can develop diabetic retinopathy. Risk of developing the eye condition can increase as a result of:

- Duration of diabetes — the longer you have diabetes, the greater your risk of developing diabetic retinopathy
- Poor control of your blood sugar level
- High blood pressure
- High cholesterol
- Pregnancy
- Tobacco use

Prevention

You can't always prevent diabetic retinopathy. However, regular eye exams, good control of your blood sugar and blood pressure, and early intervention for vision problems can help prevent severe vision loss.

If you have diabetes, reduce your risk of getting diabetic retinopathy by doing the following:

- **Manage your diabetes.** Make healthy eating and physical activity part of your daily routine. Try to get at least 150 minutes of moderate aerobic activity, such as walking, each week. Take oral diabetes medications or insulin as directed.
- **Monitor your blood sugar level.** You may need to check and record your blood sugar level several times a day — more-frequent measurements may be required if you're ill or under stress. Ask your doctor how often you need to test your blood sugar.
- **Ask your doctor about a glycosylated hemoglobin test.** The glycosylated hemoglobin test, or hemoglobin A1C test, reflects your average blood sugar level for the two- to three-month period before the test. For most people, the A1C goal is to be under 7 percent.

- **Keep your blood pressure and cholesterol under control**. Eating healthy foods, exercising regularly and losing excess weight can help. Sometimes medication is needed, too.
- **If you smoke or use other types of tobacco, ask your doctor to help you quit.** Smoking increases your risk of various diabetes complications, including diabetic retinopathy.
- **Pay attention to vision changes.** Contact your eye doctor right away if you experience sudden vision changes or your vision becomes blurry, spotty or hazy.

Remember, diabetes doesn't necessarily lead to vision loss. Taking an active role in diabetes management can go a long way toward preventing complications.

Trauma or Accidents

A 24-year-old teacher usually goes to sleep in the minibus he takes every day from his home to school. One afternoon, the minibus was bumped behind by a jeepney, shaking the passengers and bruising some of them. Unfortunately, the teacher who was fast asleep hit his head against the seat's wooden back. Since then, he complained of dizziness, severe headache and blurring vision. After thorough examination the ophthalmologist diagnosed the condition as retinal detachment. The severe blow on the head traumatized the eyeball and caused the retina to break off from it. The teacher became totally blind since then.

Another traumatic accident involved a successful blind supervisor at the department of Education. She recalled how she lost her vision at the age of 7. She was playing" espada-espadahan" with friends when one of them hit her left eye with the wooden stick. The eye became infected and she suffered from monocular blindness in the left eye. Before long, the right eye "sympathized" with the left eye and the girl became totally blind.

Rubella or German Measles

German measles (rubella) is caused by the rubella virus and spreads among humans through contact with fluids in the respiratory tract. The development (incubation) period of German measles is 14–21 days before starting to feel ill, and a rash accompanied by fever appears 1–7 days later. German measles occurs more commonly in the spring and summer months. Even in a person with a weak immune system, German measles is usually a mild illness. However, if a pregnant woman becomes infected, German measles can cause severe damage to the unborn baby.

Symptoms

Rubella is usually mild in children. Sometimes it doesn't cause any symptoms.
A pink or red-spotted rash is often the first sign of infection. It starts on the face and spreads to the rest of the body. The rash lasts about 3 days. This is why rubella is sometimes called the "3-day measles."
Along with the rash, you or your child might have:

- A mild fever -- from 99 F to 100 F
- Swollen and pink-colored eyes (conjunctivitis)
- Headache
- Swollen glands behind the ears and on the neck

- Stuffy, runny nose
- Cough
- Sore joints (more common in young women)
- General discomfort
- Lymph nodes may be swollen and enlarged

Transmission

Rubella spreads when someone who is infected coughs or sneezes tiny germ-filled droplets into the air and onto surfaces. People who catch the virus are contagious for up to a week before and a week after the rash appears.

Some people don't know they're infected because they don't have symptoms, but they can still pass the virus to others. If you've been diagnosed with rubella, tell the people who've been around you, especially any pregnant women.

Risk Factors

- Until the 1960s, rubella was a common childhood infection. Thanks to the MMR vaccine, the virus stopped spreading in the United States around 2004. Yet it still spreads in Asia, Africa, and other parts of the world. People from these areas sometimes bring the rubella virus to the United States when they travel.
- Anyone can catch rubella if they're exposed to the virus and haven't been vaccinated. Pregnant women face serious risks because rubella can cause serious complications to the baby during pregnancy.

Complications

The most serious of these could happen during pregnancy, when the virus can pass from mother to baby in the womb. The risk is highest during the first 3 months of pregnancy.

Babies who are infected can have serious birth defects called congenital rubella syndrome (CRS). This is very rare in the United States, but a baby can get it if they travel to another country where the virus spreads.

CRS is a group of health problems in a baby that can include:

- Heart defects
- Cataracts
- Deafness
- Delayed learning
- Liver and spleen damage
- Diabetes
- Thyroid problems

Some women who get rubella during pregnancy have a miscarriage. In other cases, the baby doesn't survive long after birth. It's best to get vaccinated against rubella before you get pregnant to protect your baby. You should wait at least 4 weeks after getting the vaccine to become pregnant. If you're already pregnant, you shouldn't get the vaccine.

Rubella can also cause complications in women who aren't pregnant, and in men. Young girls and women who get it can develop sore joints (arthritis). This side effect usually goes away within 2 weeks, but a small number of women will have it long term. It rarely happens in men and children.
In rare cases, rubella can cause more serious health problems, like brain infections or swelling and bleeding problems.

Prevention

The best way is to get vaccinated. Children need two doses of the MMR vaccine. They should get the first when they're between 12 and 15 months old. They should get the second between 4 and 6 years old.
Babies who'll be traveling to a country where rubella is common can get vaccinated as early as 6 months old.
If you're a woman of childbearing age and you haven't been vaccinated, get the MMR vaccine at least 1 month before you get pregnant. This is most important if you plan to travel to **countries where rubella spreads.**

Diagnosis

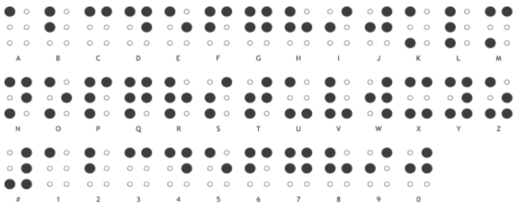
If your doctor thinks you have rubella, you may get blood tests and a virus culture to confirm that. The virus culture comes from a throat or nasal swab or from a urine sample.

Treatment

It's a virus, so antibiotics won't work.
Most of the time, the infection in children is so mild, it doesn't need to be treated. You can bring down your child's fever and ease aches with pain relievers like children's acetaminophen or ibuprofen. Don't give your child or teen aspirin because of the risk for a rare but serious condition called Reye's syndrome
If you're pregnant and think you've caught rubella, call your doctor right away. You may be able to take antibodies called hyperimmune globulin to help your body fight the virus.
As for home remedies for rubella, there aren't any that make the virus go away faster. But rest and pain relievers can help with self-care in mild cases, if needed.

Special Adaptations, Aids and technology for students Who are Blind and with Low Vision

The absence of sight leads to the reorganization of the sensory mechanisms. While vision is the major sense that gathers stimuli in the environment for processing in the brain, blindness makes the person dependent on other senses like audition or hearing, feeling or touch, olfaction or smell and gustation or taste. Persons who are blind make use of special adaptations to learn about the world that they do not see:



Braille is the primary means of literacy for blind persons. Braille is a system of reading and writing in which letters, words, numbers and other systems are made from arrangements of raised or embossed dots.

1. Braile was invented in 1830 by Louis Braille, a young blind Frenchman who played the organ in church, as means of recording church hymns and music.

The braille system is complex as shown in the picture above. A braille cell has six dots that are arranged two wide and three deep and numbered thus: The letters of the alphabet and numerals zero to ten are assigned specific to the specific combinations of the six dots. Abbreviations called “contractions” help save space and permit faster reading and writing. Some dot combinations are used to represent whole words, part-word contractions, word abbreviations, number and letter signs and others. All printed textbooks, books, music, foreign, languages, and scientific formulas can be transcribed into braille.

Blind students learn to read and write in braille by using a brailler which operates like a typewriter with six keys. The slate and stylus are another device for braille writing. It consists of a flat board to which braille paper is clipped, a braille guide where the braille dots are punched one at a time from left to right with the pointed hand-held stylus.

A portable laptop computer called VersaBraille II+ (TelesensorySystem Inc.) is available for use in doing written work, taking notes and tests.

2. Blind students use the regular **typewriter** to communicate with their teachers, classmates, and friends. Handwriting is taught so that they can sign their own names, handle bank accounts, fill up forms and vote.

3. **Manipulative and tactile aids** are used in learning mathematics, sciences, and social studies. The Cuisenaire rods with tactile markings developed by Belcastro (1989) enable the student to quickly identify by touch the different values associated with the numbers. The Cranner Abacus is used in teaching number concepts and in doing four fundamental operations. The Speech-Plus talking Calculator is a small electronic instrument that performs most of the operations of any standard calculator. It talks by” voicing” entries and results aloud and also presents them in visual form.

Blind students follow braille and verbal or taped instructions to do their lessons in social students and experiments in science. Embossed relief maps and diagrams, three dimensional models and electronic probes that give an audible signal in response to light enable blind students to participate actively in regular class activities side by side with their seeing classmates.

4. **Technological Aids** are available for blind persons. The Optacon (optical-to-tactile converter) is a small electronic device that converts regular print into a readable vibrating form for blind people. The Optacon converts print into a configuration of raised “pins’ representing the letter the camera is viewing. Through extensive training and practice, the blind person can read regular printeffectively. The blind person can also work with typewriters, calculators, computer terminals and small print.

The Kurzweil Personal reader is a sophisticated computer with an optical character recognition (OCR) system that scans and reads via a synthetic voice typeset and other printed matter.

5. **Assistive technology** enables blind persons to access to personal computers. Computer access opens to them countless opportunities for education, employment, communication and leisure activities. There are devices that magnify screen images through specialized hardware or software and computer screen-access systems that use speech recognition software to enable the user to "tell' the computer what to do.

Students who have potentially useful vision are taught to develop their ability to use their residual vision as effectively as possible. Special education teachers and other professionals are guided by certain premises about teaching students with low vision.

The Education of Students with Visual Disabilities

The integration of blind students in regular classes started in the 1960s as a component of the teacher training program for selected public-school teachers. Thus, blind boys and girls with average or better mental ability were enrolled in regular classes at the School Divisions of Pasay City, Manila and the Teacher Training Department of the then Philippine Normal College. At present, the Resources for the Blind incorporated collaborates with the Department of education in training teachers during the summer term and in mainstreaming blind and low vision students in public schools all over the country.

The special education teacher teaches skills and concepts that most children learn visually through the remaining senses: audition, touch, olfaction, gestation and other non-visual experiences. Blind students learn to read and write in **braille**. Two braille codes are learned: the Filipino Braille Code and the English Braille Code American Edition. The Philippine Printing House for the Blind at the Department of Education transcribes most of the regular textbooks and materials in braille. The books and learning materials are distributed to students who are enrolled in residential schools, special schools and regular elementary and secondary schools where they are mainstreamed. The Resources for the Blind Incorporated, a foreign-funded nongovernment organization assists the Department of Education through teacher training preparation of instructional materials and other related projects.

Blind children receive instruction in **orientation and mobility.** Orientation is the ability to establish one's position in relation to environment through the use of remaining senses. Mobility is the ability to move safely and efficiently from one point to another (Lowenfeld, 1973, cited in Heward, 2003). Instruction in orientation and mobility begins at an early age to understand the basic concepts about their bodies and their environment so they can move about effectively and safely. Then special education teacher uses appropriate methods and procedures to teach travel techniques at home, the school and the community. The long cane enables blind persons to travel independently and gain self-esteem.

The special education teacher uses manipulatives and tactile aids in teaching mathematics, science and the other subjects in the curriculum. He or she constantly plans, creates and implements activities that will enable the students to gain as much knowledge and skills as possible through their remaining senses. He or she is always on the lookout for the many opportunities for the students to participate in the regular school program. Truly, an efficient special education teacher "opens doors to a world from which his or her students who are blind or have low vision are more removed than seeing children" (Lowenfeld, 1973).

When a Student Who is Blind or has low Vision is Mainstreamed in Your Class

With the advent of inclusive education for children and youth who have disabilities, more and more students who are blind, deaf, with mental retardation, or with orthopedic impairments are enrolled in regular classes.

As a future regular teacher, you should know the basic expectations from you as a partner of the special education teacher in inclusive education. And as the child's regular classroom teacher, you will soon become sensitive to the individualized needs of the child who has a visual impairment. Your partner, the special education teacher, will be there for consultation, for special instruction in braille, mobility and orientation or travel techniques, the provision of materials adapted for the use of students.

It is important to know that children with visual impairments differ in their ability to use their remaining vision. While they rely on their other senses like audition, touch, smell, or taste, one may show preference for one sense over the others.

Have faith and trust in the ability of your blind student to participate actively in class activities. Think of ways and means to adjust highly visual lessons to tactual or auditory modes. In the absence of vision, remember that the blind student has four other sense modalities to use in gaining knowledge and learning skills in the curriculum.

Application:

How much have you learned about students who are blind or have low vision?

1. Describe how the process of vision takes place. Name and tell the functions of the parts of the visual mechanism that enables man to see.

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.

		overdone.		
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

Adapted from: *http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf*

Summary of the Lesson:

"Visual impairment" is a broad term that is used to refer to any degree of vision loss that affects a person's ability to perform the usual activities of daily life. It does not refer to the kind of vision many of us have who need to wear eyeglasses or contact lenses to read a book comfortably or see highway signs when we're driving a car. Instead, visual impairment refers to a loss of vision that cannot be corrected to normal vision, even when the person is wearing eyeglasses or contact lenses. Because it is so broad a term, "visual impairment" usually includes blindness as well.

Most visually impaired people have some usable vision. Only a small percentage of people with visual impairments are "blind"—that is, they have very little, if any, vision. Some people who are blind may have light perception or the ability to tell whether or not a light is on in a darkened room. But most will not be able to perceive light.

It's equally important to know that there is a great range of ability and disability among visually impaired people. Visual impairments range from mild to severe. And no two people see exactly alike. Even among people who have the same diagnosis, for example, retinopathy of prematurity (ROP) or retinitis pigmentosa, people may have very different visual abilities. In addition, a person's vision can be affected by daily factors and variables, such as fatigue and environmental conditions like lighting. Therefore, someone's vision can be different from day to day.

Assessment:

1. The curved transparent membrane that protects the sensitive parts of the eye.
   a. Iris c. Cornea
   b. Retina d. Pupil
2. Farsightedness
   a. Hyperopia c. Amblyopia
   b. Emmetropia d. Myopia
3. Eyeball is too long
   a. Nearsighted b. Farsighted
4. What is the purpose of acqueous humour?

a. To keep the eyeball properly inflated
b. Maintain pressure
c. Inverse the image
d. Bend the light

5. Curve transparent membrane that protects the sensitive parts of the eye
   a. Lens c. Cornea
   b. Iris d. Pupil
6. What is the iris?
   a. the colored part of the eye that controls the size of the pupil
   b. transparent jellylike fluid that fills the eye and refracts light
   c. sends messages picked up by the retina to the brain
   d. hole passes different amounts of light
7. A condition which is characterized with rapid involuntary movements of the eyeball that can result to nausea, vomiting and dizziness
   a. Cataract c. Glaucoma
   b. Diabetic retinopathy d. Nystagmus
8. Contains the eyeball, comparable in size to a ping-pong ball
   a. Retina c. Eye socket
   b. Pupil d. Cornea
9. Condition in which there is excessive pressure in the eye
   a. Coloboma c. Cataract
   b. Glaucoma d. Amblyopia
10. Cross-eyed
   a. Strabismus c. Nystagmus
   b. Amblyopia d. Cataract

Enrichment Activity:

How do students with visual disabilities manage to get an education in regular schools' side by side with their seeing classmates? What special adaptations are introduced to make mainstreaming possible?

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.

Organization	The introduction states the main topic, and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
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References/Attributions:

- Inciong, Teresita G., Quijano, Yolanda S. and Capulong, Yolanda T. Introduction to Special Education. Manila, Philippines: Rex Bookstore, 2007.